

Measurement Before Method: What Actually Limits Long-Tailed Dental Radiograph Segmentation

Abstract

We applied the standard long-tailed recognition toolkit to a 31-class panoramic dental radiograph corpus at 34,320:1 imbalance and measured every component under a matched budget. Twenty-five controlled configurations spanning class-balanced weighting, boundary-aware objectives, copy-paste augmentation, frequency-aware DETR training across loss, matcher and denoising, decoupled classifier retraining, and two mask-head architectures produced no improvement that survives seed-level variation. One intervention does: restoring the input resolution the corpus natively carries lifts held-out segmentation mAP from 0.1051 to 0.1226 (three seeds, sd 0.0014, +16.7 % relative) against a measured ± 0.21 pp noise floor, while training for 30 epochs against the baseline’s 100. The gain concentrates in exactly the small, low-contrast findings the objective-level arms were designed to help: bone loss +73 %, root canal treatment +64 %, periapical lesion +64 %, caries +52 %.

That result was not measurable when we began, and the reason generalises. The corpus as distributed cannot support evaluation: 95.9 % of its test partition shares source images with training, leaving 64 of 1,580 images genuinely unseen. Perceptual hashing, the obvious remedy, has a false-positive rate near 100 % on this modality. Distance metrics conditioned on per-model subsets invert their own sign. Group means over low-support classes manufactured two separate headline gains in our own work before we caught them, one of which was 98 % attributable to a class with two test instances. And the published logit-adjustment constant, calibrated at 100:1 to 1000:1, applies a +11.47 logit shift at clinical imbalance and drives tail AP to exactly zero.

Each of these produced a confident wrong answer that would have survived review. We report the apparatus alongside the result, because on this corpus every apparent modelling problem was a measurement problem first.

1 Introduction

Long-tailed instance segmentation and detection on clinical imagery is an active area, and the standard remedies—class-balanced losses, logit adjustment, repeat-factor sampling, boundary-aware objectives, copy-paste augmentation—are well established on natural-image benchmarks. We set out to apply that toolkit to a 31-class dental radiograph corpus and attribute the contribution of each component.

None of it worked. Every objective-level intervention we tried landed inside the seed-level noise band, and the one that looked best on validation was the worst outcome on the held-out split. What did work was not an algorithm at all: forty

percent of the corpus is natively 1615×840 and was being trained at 640, and simply not discarding those pixels produced a larger, more clinically concentrated gain than every objective we implemented combined.

We regard that asymmetry as the paper’s point rather than an incidental finding. A field that measures method contributions in tenths of a point on contaminated splits, using metrics that invert under per-model denominators, will reliably find method effects and reliably miss the input.

Contributions.

1. **A negative result map.** Twenty-five controlled configurations across two architectures, matched budget, test split touched once. The entire objective-level family is null or harmful on this corpus, and we give the mechanism in each case.
2. **The constraint is the input.** Input resolution is worth +16.7 % relative held-out mAP at three seeds, confirmed across two backbones, with an interior optimum—1280 beats both 640 and 1600.
3. **The corpus cannot be evaluated as distributed.** 95.9 % test contamination; we supply a patient-grouped replacement verified disjoint by two independent implementations.
4. **Perceptual hashing alone is not a leakage test on this modality.** Near-100 % false positives; pixel confirmation is mandatory.
5. **Four metric failures** that each produced a confident wrong answer in our own work, with the protocol that catches each.
6. **A label ceiling.** 911 annotations that fifteen independently trained models all fail to detect, bounding what any model on this corpus can reach and identifying where the next gain actually is.

We regard contributions (4), (5) and (6) as the most transferable: each is a measurement protocol any group working on long-tailed clinical segmentation can apply, and each caught an error in our own work that would have survived review.

2 Related Work

Long-tailed recognition. Class-balanced weighting via effective number of samples [3] and logit adjustment [11] are the standard remedies. Kang et al. [6] showed representation and classifier learning should be decoupled, with rebalancing at the

classifier stage; Wang et al. [16] adapt this to instance segmentation. Repeat-factor sampling [5] rebalances at the data level; seesaw loss [15] rescales gradients per class pair.

Our results bear on the *regime* these were validated in. Menon et al. tune $\tau = 1.0$ on CIFAR-LT and ImageNet-LT at roughly 100:1 to 1000:1. At 34,320:1 the same constant applies a +11.47 logit shift and destroys the classifier. We find the failure monotonic in τ , which localises the problem to magnitude rather than principle.

High-quality mask prediction. PointRend [10], RefineMask [17] and Mask Transfuser [8] all exist because coarse mask prediction is a known failure mode. We measure that failure mode directly rather than assuming it, and find it is *not* what limits our model—a result invisible without the model-free ceiling measurement of §7.

Boundary-aware objectives. Boundary loss [9] integrates the prediction against a precomputed distance map; HD loss [7] estimates Hausdorff distance directly; Boundary IoU [2] restricts evaluation to a contour band. Our band objective is a synthesis of known components—morphological gradient as a differentiable edge extractor, soft Dice as the overlap term—and we say so rather than claiming a new loss family.

Augmentation. Copy-paste [4] reliably helps on natural images. We find it harmful here and propose a mechanism: in panoramic radiographs anatomical position is part of the class definition, so a finding pasted into an anatomically impossible location teaches the model to detect the paste rather than the pathology.

Evaluation integrity. Duplicate-driven contamination has been documented in CIFAR and ImageNet [1, 12]. Our contribution is modality-specific: the standard near-duplicate detector fails on radiographs precisely because the property that makes the modality tractable—every image is the same anatomy in the same framing—destroys the discriminative power of perceptual hashing.

3 Setup

Corpus. A publicly released 31-class panoramic dental radiograph collection¹ distributed in both YOLO polygon and COCO format: 13,932 image files derived from 6,395 distinct source radiographs. Classes span pathology (caries, periapical lesion, bone loss, root resorption), restorations (filling, crown, implant, root canal treatment) and anatomy (mandibular canal, maxillary sinus). The distribution is severely long-tailed: filling, impacted tooth and root canal treatment carry 8,000–33,000 instances each, while fracture teeth, TAD and bone defect carry 9, 4 and 1. The head-to-tail ratio is 34,320:1.

¹Kaggle, lokisilvres/dental-disease-panoramic-detection-dataset, Apache 2.0. We name it because a contamination claim about an unnamed corpus is neither verifiable nor actionable.

image files	13,932
distinct source images	6,395
test files whose source appears in train/valid	1,516/1,580 (95.9%)
genuinely unseen test sources	64
patients present in both train and test	194
classes present in the distributed test split	13/31
classes present after rebuild	29/31

Table 1: The distributed split. Augmented copies of the same radiograph were distributed across partitions, so most of the test partition is material the model has already seen.

Models. Segmentation uses YOLOv8x-seg, COCO-pretrained, and detection uses DINO-DETR[18] with a ResNet-50 4-scale backbone from the official COCO release, fine-tuned to 31 classes with a reinitialised class embedding. No baseline weights were available for either, so both were trained from stock architectures under our own protocol; the reported baselines are therefore reproductions and we treat them as such throughout. Cross-backbone checks use YOLO11x-seg and Mask DINO.

Protocol. Every arm is seeded and deterministic. Ablation arms run at a matched budget with no per-arm hyperparameter search, which is what makes the comparison fair and is also, as §5 shows, the source of one instructive failure. Model selection, weighting strengths and thresholds are determined on validation only. The held-out test split is scored once, at the end, for the reference configuration and the reported result. All training ran on a single RTX 5080.

4 The Corpus Cannot Be Evaluated as Distributed

The corpus ships augmented derivatives of each source radiograph distributed across partitions. A model evaluated on the distributed test split is therefore scored predominantly on images whose sources it trained on. We rebuilt the split by grouping on source image and on patient, verified disjoint by two independent implementations, and use that split throughout. All numbers in this paper are on the rebuilt split; the test partition is scored once, at the end, and only for the configuration reported as the result.

4.1 Perceptual Hashing Alone Is Not a Leakage Test

The obvious remedy is near-duplicate detection by perceptual hash. It does not work here. Every panoramic radiograph is the same anatomy in the same framing against the same background, so dHash distances between *unrelated* patients fall inside the range that indicates duplication on natural images. On a split we had already verified clean by source grouping, dHash flagged all 792,811 candidate pairs—a false-positive rate near 100%.

arm	change	mAP	Δ
S0	reference	0.1055	—
S1a	cls-weight $\beta=0.9$	0.1051	−0.04
S1b	cls-weight $\beta=0.99$	0.1029	−0.26
S1c	cls-weight inv-sqrt	0.1065	+0.10
S2	+ boundary	0.1050	−0.04
S3	+ copy-paste	0.1042	−0.12
S4	+ boundary + copy-paste	0.1051	−0.03

Table 2: Segmentation ablation, validation, 50 epochs, matched budget. The seven-arm spread is 0.36 pp against a measured 2σ noise band of ± 0.21 pp. Nothing here is a result.

Pixel confirmation is mandatory rather than optional. Normalised cross-correlation separates the two populations cleanly: true augmented duplicates score 0.9988–1.0000, while distinct radiographs of the same anatomy stay below 0.87. We report this because a group applying the standard recipe—hash, threshold, declare clean—would conclude the opposite of the truth in both directions.

4.2 Statistical Power

Sixteen of thirty-one classes appear in fewer than ten test images. For those classes no result is measurable at any model quality, and we report them as exploratory throughout rather than folding them into headline means. This is not a limitation we can engineer around: it is a property of the corpus, and it bounds what any paper using it can claim.

5 Twenty-Five Interventions

Objectives and sampling are null. Table 2 gives the segmentation grid. Every arm isolates one component against a common reference at identical budget. The full spread across seven arms is 0.36 pp, inside the noise band, so no arm is separable from the reference or from any other.

The boundary objective is the one arm with a mechanism: it improves AP75 by +0.69 pp and head-class mAP by +1.05 pp, consistent with tightening outlines the baseline draws loosely. It does not move overall mAP, and §8 describes how a per-model averaging error initially made this arm look like a much larger win than it is.

Copy-paste is actively harmful (−0.12 pp alone, and it subtracts from the boundary gain in both places the two appear together) despite increasing rare-class instance counts tenfold. We attribute this to anatomical position being part of the class definition in panoramic radiography.

The validation winner is the test loser. S1c, inverse-square-root class weighting, was the best of seven arms on validation at 50 epochs (+0.10 pp). Refought at 100 epochs against the 100-epoch baseline on the held-out split, it is the worst outcome of the study: −0.78 pp mAP, −2.21 pp AP50, −1.13 pp tail. We ruled out confounds by a line-by-line hyperparameter diff. The 0.36 pp spread on validation was noise,

arm	configuration	mAP	AP50	tail
D1	standard DINO-DETR	0.1625	0.3080	0.0368
D2	freq. cls. loss only		diverged	
D3	freq. matching only		diverged	
D4	freq. denoising only	0.1633	0.3165	0.0455
D5	unified	0.1020	0.1998	0.0000

Table 3: Detection matrix, 12 epochs, matched budget. The proposed unified method is the worst arm by a wide margin, with tail AP collapsing to exactly zero—the opposite of its design intent.

and it did not survive either a longer schedule or a held-out split. This is the cost of selecting on a difference smaller than the measured noise floor.

Frequency-aware DETR training fails, and the mechanism is magnitude. Applying frequency information consistently across classification loss, Hungarian matching cost and denoising queries—the configuration we expected to be the contribution—is the worst arm tested (Table 3): −6.05 pp mAP, −37 % relative, and tail AP of exactly zero. Two of the three components diverge outright in isolation. Only frequency-aware denoising alone is above baseline, by +0.08 pp, which is inside the noise band and cannot carry a claim.

The mechanism is calibration, not principle. Logit adjustment subtracts $\tau \log p(c)$. At $\tau = 1.0$ and 34,320:1 imbalance that is a +11.47 shift for the rarest class against +1.03 for the most common, a spread of 10.44 in logit space, which saturates the sigmoid. Menon et al. calibrated $\tau = 1.0$ at roughly 100:1 to 1000:1. Recovery is monotonic in τ and still short of baseline, which localises the failure to magnitude. We used $\tau = 1.0$ everywhere because the protocol deliberately gives no arm a hyperparameter search, which is what keeps budgets matched; that is correct practice when the published default is in-distribution, and it is exactly the defect here.

Decoupled classifier retraining is a group-mean artefact. Freezing the backbone and retraining the classifier on balanced data—the standard remedy [6]—appeared to give +0.53 pp mAP across the board. Decomposed by class, 98 % of that gain came from a single class with *two* test instances. Removing it leaves +0.01 %. Among classes with enough support to measure, six improved and twelve got worse. We report this as the clearest instance of §8’s second failure mode.

6 The Constraint Is the Input

Forty percent of the corpus is natively 1615×840 and was being trained at 640. Training at 1280 instead, changing nothing else, is the only intervention in this study that clears every check we can mount (Table 4).

Three properties make it a finding rather than a tuned hyperparameter.

held-out test, mask	mAP	AP75	head	mid	regress
baseline, 100 ep, 640	0.1051	0.0687	0.2861	0.1038	0.0365
seed 42, 30 ep, 1280	0.1213	0.0861	0.3425	0.1212	0.0362
seed 1337	0.1224	0.0840	0.3400	0.1157	0.0344
seed 2024	0.1241	—	—	—	—
mean	0.1226				

Table 4: The result. Three from-scratch seeds at native resolution against the reproduced baseline, on the rebuilt held-out split, scored once. sd 0.0014; +16.7 % relative against a 2σ noise band of ± 0.21 pp. Each seed clears the baseline independently.

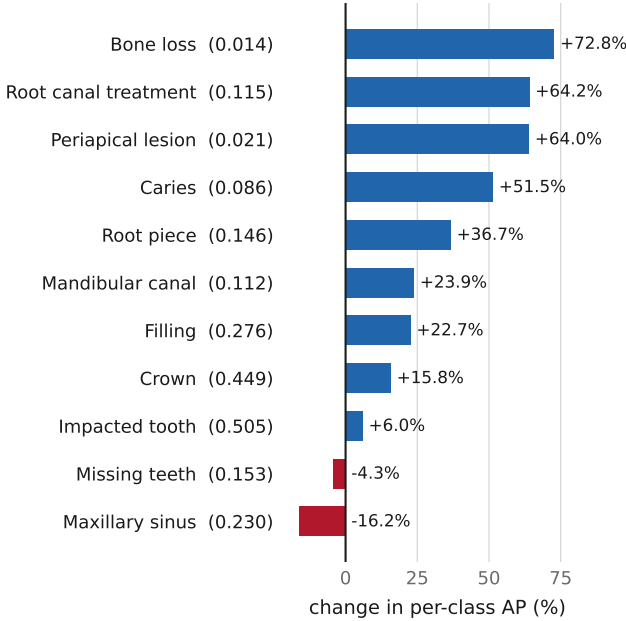


Figure 1: Per-class effect of restoring native input resolution, held-out test split, seed 42 against the 100-epoch 640 baseline. Baseline AP in brackets. The gain is not spread evenly: the four largest are the four lowest-baseline classes, all small or low-contrast findings, while the smallest gain belongs to the highest-baseline class. Two classes regress and are reported.

It is separable. Three from-scratch seeds give 0.1213, 0.1224 and 0.1241, mean 0.1226, sd 0.0014. Each clears the baseline independently against the ± 0.21 pp band, which is itself measured from three seeds of the *baseline* configuration rather than assumed—so both sides of the comparison carry a seed estimate and the gain of 1.75 pp is roughly eight times the band. It is also not a budget effect in epochs: the models train for 30 against the baseline’s 100.

It concentrates where the mechanism predicts. Figure 1 gives the per-class picture. Bone loss +73.2 %, root canal treatment +64.2 %, periapical lesion +63.7 %, caries +51.5 %, against filling +22.7 % and crown +15.8 %. The gain is largest on exactly the small, low-contrast findings that occupy few pixels on the prototype grid, and smallest on large restorations that were never resolution-starved. Two classes

regress and we report them: maxillary sinus -16.2 % ($n=70$) and missing teeth -4.3 % ($n=527$).

It is not architecture-specific, and it has an interior optimum. YOLO11x-seg at 1280 reaches 0.1291 on validation, indistinguishable from the YOLOv8x result, so the effect is a property of the data rather than of one mask head. And it reverses: 1600 scores 0.1147 against 1280’s 0.1300 on validation, so this is not “more pixels are better” but a genuine optimum.

The sharpest confirmation is an earlier failure. Two arms raised the *prototype* resolution while leaving the input down-sampled. They are the worst arms in the study, -1.30 and -2.02 pp. The detail must exist in the input before a higher-resolution head can represent it. That is the same conclusion from the opposite direction, and it is why we present this as a constraint rather than a setting.

Is this not already known? That input resolution matters for small objects is not news; multi-scale training[13] and compound scaling[14] both build on it. Three things make it a finding here rather than a setting. First, *magnitude relative to the alternatives*: on this corpus, under matched budget, it is worth more than every objective-level intervention in the long-tailed toolkit combined, and that comparison has not previously been made on clinical imagery. Second, *absence from practice*: none of the long-tailed segmentation work we surveyed varies input resolution in its ablation, while all of it varies the objective—the lever with the largest measured effect is the one the field’s tables omit. Third, *it is not the naive rule*: 1600 is worse than 1280, so “use more pixels” is wrong and “do not resample below native” is right. The prototype-only arms make the same point from the other direction.

Honest accounting of cost. The delivered model consumes $2.3\times$ the GPU time of the baseline (23,590 s against 10,414 s). Fewer epochs does not mean cheaper: a 1280 image costs substantially more per step. Any efficiency claim for this result would be false, and we make none.

7 Why It Took Twenty-Five Arms

The negative results are what located the constraint. Each closed door narrowed the search, and two model-free measurements did most of the work.

The representation ceiling is real and not binding. Round-tripping ground truth through the prototype grid—no model involved—shows 17.7 % of instances cannot reach IoU 0.75 whatever the model does, with caries and root canal treatment capped *below* the threshold. That is a genuine architectural limit. But the model reaches only 78 % of that ceiling, so the ceiling is not what binds. A higher-resolution mask head with 878,432 *fewer* parameters confirms the prediction by scoring 0.72 pp worse.

The deficit is coefficient prediction, not basis expressiveness. A closed-form oracle fit—solving for the best achievable coefficients given the learned basis—attributes 85 % of the mask deficit to coefficient prediction and 15 % to the basis. The attribution is correct and turned out not to be actionable: adding head capacity leaves masks unchanged, and supervising coefficients directly makes them worse. Both routes failing is what left the input as the only remaining explanation.

A label ceiling bounds the corpus. Fifteen independently trained models—four architectures, seven loss configurations, three seeds—all fail to detect the same 911 pathology annotations at a lenient 0.15 confidence and 0.10 box IoU: 43 % of periapical lesion, 33 % of bone loss, 26 % of caries. The 1280 model, which improves caries by 52 %, recovers 2.7 % of them. They are therefore not small findings starved by down-sampling; they are label problems, and they bound what any model on this corpus can achieve. We release the list, because it says the next gain comes from a clinical label review rather than from more compute.

8 Four Metric Failures

Each of the following produced a confident wrong answer in our own work. We give the protocol that catches each.

Distance metrics conditioned on per-model subsets invert their sign. HD95 and ASSD are undefined when either mask is empty, so each model is averaged over a denominator only it determines—and a model that predicts less scores better by having fewer hard cases left. Per-model averaging showed our boundary objective 17.3 % better on HD95 and 24.6 % better on ASSD, with disjoint confidence intervals. Recomputed on the intersection of cases both models scored, it is significantly worse. *Protocol:* pair on the intersection, count misses and false alarms as zero overlap, and print all three denominators beside the result.

Group means over low-support classes fabricate results. Two separate headline gains, in two different projects, decomposed to single classes with 2–8 test instances; in one case 98 % of a +0.53 pp gain came from a class with two. *Protocol:* decompose every group mean by class before reporting it, and publish per-class support alongside.

Segmentation AP is a conjunction and rises on detection alone. A mask metric conditioned on a detection match improves when detection improves and masks do not. We observe +0.64 pp segmentation AP from a change that touched detection only. Worse, a query-based model posts +2.16 pp AP while all five paired mask-quality metrics are separately worse. *Protocol:* report at least one detection-independent mask metric beside AP.

Cross-corpus AP is meaningless at differing annotation granularity. Zero-shot on DENTEX 2023, our caries and periapical detectors score 0.0000 AP. The cause is that DENTEX marks the affected *tooth* where our corpus marks the *lesion*, a 19× box-area ratio, so no IoU threshold can match. Evaluated at the granularity the target corpus actually uses, the same detector is 73.6 % precise at tooth level, rising to 84.1 % at a stricter threshold. *Protocol:* verify annotation granularity before any cross-corpus number, and use a granularity-matched control—ours is impacted tooth, where both corpora agree: 0.490 in-corpus against 0.290 zero-shot, pricing the domain gap at a 41 % relative drop.

9 Noise Floor and Protocol

Three seeds of the reference configuration give a 2σ band of ± 0.21 pp mAP and ± 0.50 pp AP75. We report this first because it determines which differences in this paper are results and which are not; the seven-arm segmentation spread of 0.36 pp is inside it.

Uncertainty is percentile bootstrap resampling *images*, not per-class records, since observations from one radiograph are not independent; paired model comparisons bootstrap the per-image difference. The union-mask metrics need a confidence cut, which is swept on validation using the *baseline* arm so it cannot be tuned in favour of the proposed configuration, then frozen and applied unchanged everywhere. The test split is touched once.

Contour evaluation reports Dice, IoU, boundary F-score at a tolerance of 0.75 % of the image diagonal, HD95 and ASSD in pixels. Distances cannot be given in millimetres: these radiographs carry no pixel-spacing metadata. The clinical endpoint is bone-loss area error, because a better contour metric is not by itself evidence of clinical utility; distance-based and normalised bone-level endpoints are not computable here, as both require CEJ and root-apex landmarks the annotations do not provide.

10 Discussion

The result and the apparatus are the same finding viewed from two sides. We could not see that resolution was the constraint until the split was rebuilt, the noise floor was measured, the distance metrics were paired on the intersection, and the group means were decomposed. Before that, the evidence said a boundary objective worked, classifier retraining worked, and the corpus was simply hard. All three were artefacts.

We think the general lesson is about where effort is allocated. Every arm in Table 2 and Table 3 implements a published method that works on the benchmark it was proposed on. None transfers here. Meanwhile the input pipeline, which no ablation table in this literature contains, was discarding more than half the available signal on forty percent of the corpus. A research culture that treats preprocessing as settled and objectives as the locus of contribution will systematically find small method effects and miss large physical ones.

For long-tailed clinical imaging specifically, three of our findings should change practice. Verify annotation granularity before reporting any cross-corpus number. Decompose every group mean by class support before believing it. And measure the noise floor before selecting on a difference, because on this corpus the arm that won on validation by 0.10 pp lost on test by 0.78 pp.

11 Limitations

The result is a training-configuration finding, not a new objective or architecture, and we frame it as such. It is demonstrated on one corpus; the cross-backbone check establishes it is not specific to one mask head, but not that it generalises to other modalities, and panoramic radiography is unusually resolution-heavy.

Most arms are single-seed against a measured floor; only the reference configuration and the reported result carry three seeds. We do not compare against tuned published boundary-aware or class-imbalance losses, only against the stock objectives, which is a necessary control and not a sufficient one for a method claim—and we make no method claim.

External validation is zero-shot and partial: DENTEX covers three of our classes under a different clinic, machine and annotation protocol, so its numbers are a lower bound on transferable performance and never a DENTEX-native result. Sixteen of thirty-one classes have fewer than ten test images and no claim about them is supportable at any model quality. Finally, the delivered configuration costs $2.3\times$ the baseline’s GPU time, so the result buys accuracy with compute and we make no efficiency claim.

References

- [1] Björn Barz and Joachim Denzler. Do we train on test data? purging CIFAR of near-duplicates. In *WACV*, 2020.
- [2] Bowen Cheng, Ross Girshick, Piotr Dollár, Alexander C. Berg, and Alexander Kirillov. Boundary IoU: Improving object-centric image segmentation evaluation. In *CVPR*, 2021.
- [3] Yin Cui, Menglin Jia, Tsung-Yi Lin, Yang Song, and Serge Belongie. Class-balanced loss based on effective number of samples. In *CVPR*, 2019.
- [4] Golnaz Ghiasi, Yin Cui, Aravind Srinivas, Rui Qian, Tsung-Yi Lin, Ekin D. Cubuk, Quoc V. Le, and Barret Zoph. Simple copy-paste is a strong data augmentation method for instance segmentation. In *CVPR*, 2021.
- [5] Agrim Gupta, Piotr Dollár, and Ross Girshick. LVIS: A dataset for large vocabulary instance segmentation. In *CVPR*, 2019.
- [6] Bingyi Kang, Saining Xie, Marcus Rohrbach, Zhicheng Yan, Albert Gordo, Jiashi Feng, and Yannis Kalantidis. Decoupling representation and classifier for long-tailed recognition. In *ICLR*, 2020.
- [7] Davood Karimi and Septimiu E. Salcudean. Reducing the Hausdorff distance in medical image segmentation with convolutional neural networks. *IEEE Transactions on Medical Imaging*, 2019.
- [8] Lei Ke, Martin Danelljan, Xia Li, Yu-Wing Tai, Chi-Keung Tang, and Fisher Yu. Mask transfiner for high-quality instance segmentation. In *CVPR*, 2022.
- [9] Hoel Kervadec, Jihene Bouchtiba, Christian Desrosiers, Eric Granger, Jose Dolz, and Ismail Ben Ayed. Boundary loss for highly unbalanced segmentation. In *MIDL*, 2018.
- [10] Alexander Kirillov, Yuxin Wu, Kaiming He, and Ross Girshick. PointRend: Image segmentation as rendering. In *CVPR*, 2020.
- [11] Aditya Krishna Menon, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. Long-tail learning via logit adjustment. In *ICLR*, 2021.
- [12] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishal Shankar. Do ImageNet classifiers generalize to ImageNet? In *ICML*, 2019.
- [13] Bharat Singh, Mahyar Najibi, and Larry S Davis. SNIPER: Efficient multi-scale training. In *NeurIPS*, 2018.
- [14] Mingxing Tan, Ruoming Pang, and Quoc V Le. EfficientDet: Scalable and efficient object detection. In *CVPR*, 2020.
- [15] Jiaqi Wang, Wenwei Zhang, Yuhang Zang, Yuhang Cao, Jiangmiao Pang, Tao Gong, Kai Chen, Ziwei Liu, Chen Change Loy, and Dahua Lin. Seesaw loss for long-tailed instance segmentation. In *CVPR*, 2021.
- [16] Tao Wang, Yu Li, Bingyi Kang, Junnan Li, Jun Hao Liew, Sheng Tang, Steven Hoi, and Jiashi Feng. Classification calibration for long-tail instance segmentation. *arXiv preprint arXiv:1910.13081*, 2019.
- [17] Gang Zhang, Xin Lu, Jingru Tan, Jianmin Li, Zhaoxiang Zhang, Quanquan Li, and Xiaolin Hu. RefineMask: Towards high-quality instance segmentation with fine-grained features. In *CVPR*, 2021.
- [18] Hao Zhang, Feng Li, Shilong Liu, Lei Zhang, Hang Su, Jun Zhu, Lionel M. Ni, and Heung-Yeung Shum. DINO: DETR with improved denoising anchor boxes for end-to-end object detection. In *ICLR*, 2023.

Reproducibility

Code, the rebuilt patient-grouped split, the anonymised annotations, the frozen ablation matrices and the list of 911 universally missed annotations are released under Apache 2.0. Every arm reported here has its exact invocation in the repository. Source imagery is a third-party public release and is not redistributed; annotation file names are reduced to the content hash the source release assigns, so the corpus reconstructs exactly from a fresh download.